

Confidence Before Competence: Entity Frequency and Calibration During Language Model Pretraining

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Abstract

A language model that gets a fact wrong is only dangerous if it sounds sure. We ask how the number of times a fact appeared in pretraining affects the match between a model’s confidence and its accuracy, and when during pretraining that relationship forms. We use LMEnt (Gottesman et al., 2026), which releases language models together with the entity-annotated corpus they were trained on, so the number of times a subject and its answer co-occurred is known rather than estimated. Scoring 2,993 PopQA facts on 61 checkpoints across four models, we find that a randomly initialised model already reports a confidence-accuracy gap of about 0.13, and that training closes it in proportion to exposure: by 0.28 for facts with 61 to 200 co-occurrences, while leaving it unchanged for facts whose entities never co-occur. The ordering holds under all three scoring functions we tried, though the levels do not. The gap is already present at step 1,000 of 658,032, where confidence varies far less across frequency than accuracy does: across the ten bins accuracy rises by 0.275 and confidence by 0.076. Neither model scale from 170M to 1B, nor six epochs instead of one, nor rephrased prompts remove the effect. When the model is wrong it tends to fall back on the most likely answer for the relation, and it does so more often for rare entities. Our code, probe set and per-checkpoint results are released.

1 Introduction

Language models store a lot of facts, and they store them unevenly. Facts about widely discussed entities are recalled well; facts about rare ones are not (Mallen et al., 2023; Kandpal et al., 2023). Less is known about what the model does when recall fails. It could spread probability evenly over the options, which would tell a user that it does not know, or it could commit to one wrong answer, which would not.

This paper measures which of those happens, as a function of how often the fact appeared in pretraining and how far training has progressed. Both questions need something that is normally unavailable: the training corpus of the model under test, annotated well enough to count how often two entities appeared together. LMEnt (Gottesman et al., 2026) provides exactly that, along with intermediate checkpoints, so we can count occurrences instead of approximating them with a popularity proxy.

We score 2,993 PopQA facts as ten-way multiple choice at each checkpoint and record two numbers per fact: whether the model picks the gold answer, and how much probability it puts on its choice. Averaged over facts with the same co-occurrence count, the difference between the two is a simple measure of how far confidence runs ahead of knowledge.

Three findings follow. First, that difference tracks co-occurrence count closely, and it is not present in a randomly initialised model, so training produces it. Second, it appears very early: by step 1,000 of 658,032 the model is about equally confident about facts it has never seen evidence for and facts it has seen hundreds of times, even though its accuracy on the two groups already differs by a factor of two and a half. Confidence arrives before the knowledge it is supposed to reflect. Third, the effect survives every control we ran, including a six-fold increase in model size.

2 Background

Frequency and factual recall. Kandpal et al. (2023) entity-link five pretraining corpora and count documents containing both the question and answer entities, then show strong correlational and causal relationships between that count and question-answering accuracy. They also estimate that models would need to be scaled by many or-

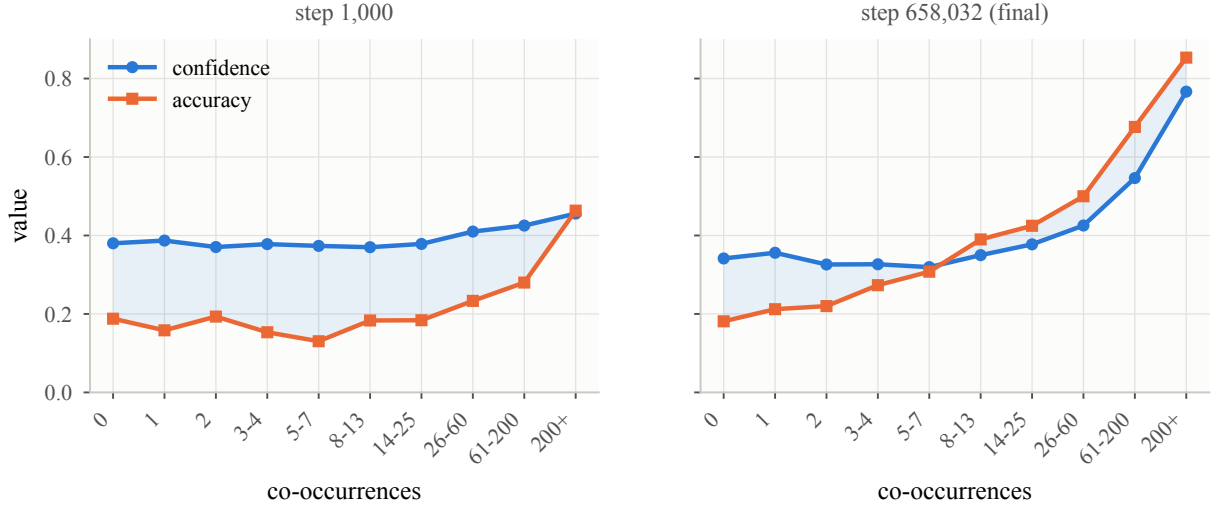


Figure 1: Confidence and accuracy by co-occurrence count, early in training and at the end. At step 1,000 confidence is nearly flat across frequency while accuracy is not. The shaded region is the gap between them.

ders of magnitude before they answer well on questions with little support in pretraining. [Mallen et al. \(2023\)](#) introduce PopQA, 14k questions about long-tail entities, and report that scaling mainly improves recall of popular knowledge and does little for the tail. Both papers measure accuracy. Neither asks what the model’s confidence does.

Knowledge during pretraining. [Chang et al. \(2024\)](#) study acquisition directly by injecting facts a model has not seen into intermediate checkpoints and tracking how the probability of those facts rises and then decays. They report a power-law relation between training steps and forgetting, faster forgetting when training data is duplicated, and greater robustness at larger batch sizes. Because they inject facts, their design controls exposure precisely but does not measure facts the model met naturally. We take the complementary route and observe naturally occurring facts whose corpus counts are known.

LMEnt. LMEnt ([Gottesman et al., 2026](#)) releases a Wikipedia-based pretraining corpus annotated with entity mentions, an entity-based retrieval method over that corpus, and pretrained models of up to 1B parameters with intermediate checkpoints. Among several candidate predictors of model accuracy they find that Subject-Answer Chunks, defined as the number of corpus chunks in which the subject and answer entities co-occur, correlates most strongly with performance, better than Wikipedia pageview popularity. They also analyse knowledge acquisition over training, evaluating their 1B model every 20K steps and marking a fact as learned when

the model answers at least one question linking the subject to the answer. They report that rates of both learning and forgetting rise with fact frequency, and close by noting that the interplay between internal mechanisms and training dynamics is not yet fully understood. Their analysis is framed in terms of whether a fact is known. We use the same resource to ask a different question about the same checkpoints, namely how sure the model is.

Calibration. A calibrated classifier assigns confidence equal to its probability of being correct. [Guo et al. \(2017\)](#) show that modern neural networks are poorly calibrated and summarise miscalibration with expected calibration error ([Naeini et al., 2015](#)). That statistic is unsigned. We report the signed difference between mean confidence and mean accuracy instead, because the direction is what we are after: a model that is too sure and one that is not sure enough are different failures. Scoring multiple-choice options by raw string probability is also known to be unreliable, because surface forms compete for probability mass ([Holtzman et al., 2021](#)). Section 3.3 shows this matters here. Closer to our setting, [Desai and Durrett \(2020\)](#) find that fine-tuned BERT and RoBERTa classifiers are calibrated in-domain and degrade out-of-domain, while staying better calibrated than models trained from scratch, and [Kadavath et al. \(2022\)](#) report that large models are well calibrated on multiple-choice questions when the options are presented in a suitable format, and can partly predict whether they know an answer at all. Both study fully trained models. We ask when that calibration arrives dur-

Co-occurrences	Available	Sampled	Test
0	2450	298	85
1	2307	297	83
2	1586	300	75
3–4	1999	300	61
5–7	1556	299	83
8–13	1437	300	70
14–25	985	299	70
26–60	769	300	90
61–200	522	300	65
200+	656	300	75
Total	14267	2993	757

Table 1: Facts per co-occurrence bin. Splits are 50/25/25 over 2,826 distinct subject entities, so no entity appears in more than one split. Only the prediction task in Section 4.5 uses the splits. Everything else is measured over all 2,993 facts.

ing pretraining, and whether it holds for facts the corpus barely contains.

3 Experimental Setup

3.1 Facts and frequency

We use the PopQA subset released with LMEnt, which pairs each question with the identifiers of the chunks where the subject entity, the answer entity, and both together appear. The co-occurrence count is the paper’s Subject-Answer Chunks measure. We sort the 14,267 facts by that count and split them into ten value-based bins, which therefore differ in size, then sample 300 facts per bin, giving 2,993 facts after discarding a handful for which we could not build a full candidate set. Bin edges and counts are in Table 1.

Two groups deserve mention. In the lowest bin the subject and answer never co-occur, so the corpus contains no direct evidence for the fact. For 112 of those facts the subject entity is absent from the corpus altogether, a stricter control we report separately.

3.2 Probes

Each of the 16 PopQA relations gets one hand-written cloze template, for example *The director of {subject} is.* Templates are fixed in advance and applied mechanically, so probe construction is reproducible and involves no model in the loop.

For each fact we build a ten-option multiple choice question: the gold answer plus nine distractors drawn from the objects observed for the same relation, so all options are type-consistent. Distractors are matched to the gold on token length,

which limits the length bias described below, and screened against the gold’s alias list so no distractor is silently correct. Mean gold length is 2.95 tokens against 2.93 for distractors. Chance accuracy is 0.100.

3.3 Scoring

We score each option by its length-normalised log probability under the prompt and take a softmax over the ten options. Accuracy is whether the gold option ranks first; confidence is the probability of the top-ranked option.

The choice of scoring function decides what confidence even means here. At step 0 the model’s weights are random, so whatever confidence a scorer reports there is floor rather than belief. The three disagree sharply: summed log probability reports 0.439, because it ranks options largely by how short they are, domain-conditional PMI (Holtzman et al., 2021) reports 0.396, and length-normalised scoring reports 0.216. We use length-normalised scoring throughout, because the lowest floor leaves the most room to detect confidence that reflects knowledge, and we report the other two alongside it.

We also record a free-generation probe: greedy decoding of ten tokens from the same prompt, scored by normalised string match against the answer aliases, with the mean token log probability of the generated span as its confidence.

3.4 Models

We evaluate the LMEnt models at 170M, 600M and 1B parameters trained for six epochs, and the 1B model trained for one epoch. Of the checkpoints LMEnt describes, we use those publicly released on HuggingFace: 25 for the 1B six-epoch model, including every 1,000 steps over the first 10,000, and 12 each for the others. That gives 61 checkpoint evaluations, plus six more for the template controls, two alternative wordings at three checkpoints of the 1B model. All inference runs in float32, because the 1B model’s activations reach roughly 3×10^6 , far above the float16 range, and float16 produces non-finite scores on every trained checkpoint. Total compute is about 8 GPU-hours on Tesla T4s.

The probe set is fixed by a seed and identified by the hash 5132e8b6ffaa40cb. Every run verifies it before scoring. We release the notebooks, the 2,993-fact probe set, and per-fact results for all 67 runs, along with a manifest per run recording the GPU, numeric precision and library versions used.

4 Results

4.1 Confidence outruns accuracy for rare facts

Figure 1 shows confidence and accuracy against co-occurrence count at step 1,000 and at the end of training. At step 1,000 the two curves are almost independent: accuracy already rises by 0.275 across the bins while confidence rises by 0.076. Over the seven rarest bins both are flat, at 0.130 to 0.193 and 0.370 to 0.387. The model is then close to equally sure about facts it has no evidence for and facts it has seen a few dozen times, while being wrong about most of both.

By the end of training the curves have partly converged, but a gap remains at the rare end. Figure 2 plots the difference. It falls steadily from +0.160 in the lowest bin to -0.130 in the second-highest, crossing zero between 5–7 and 8–13 co-occurrences. Beyond that point the model is underconfident: it is right more often than it claims. Where the crossing falls, and whether it happens at all, depends on the scoring function. Under PMI no bin is negative. The decline across frequency is the same under all three (Table 2). Rank correlation between log co-occurrence count and the per-fact gap is -0.253 for length-normalised scoring, -0.287 for summed and -0.206 for PMI, with $p < 10^{-29}$ in every case.

The same measurement on the randomly initialised checkpoint is flat, between 0.100 and 0.156, with no ordering by frequency. That floor is not zero, because the largest of ten softmax probabilities exceeds 0.100 by construction. Measured against it, training closes the gap by 0.284 for facts with 61 to 200 co-occurrences, while leaving it statistically unchanged for facts whose entities never co-occur: the gap there moves by +0.009, and the initial value of 0.151 sits inside the final bootstrap interval of $[0.117, 0.202]$. The change is almost perfectly ordered by frequency (Spearman -0.98). The model does not learn to be overconfident about rare facts. It begins overconfident about everything, and exposure is what corrects it, except where there is nothing to correct it with.

4.2 Confidence settles early, accuracy does not

Pooled over all facts, confidence moves from 0.216 at initialisation to 0.393 at step 1,000 and then spans 0.038 across the remaining 657,000 steps, a net change of 0.021, while accuracy rises by 0.187 over the same span. That pooled summary hides

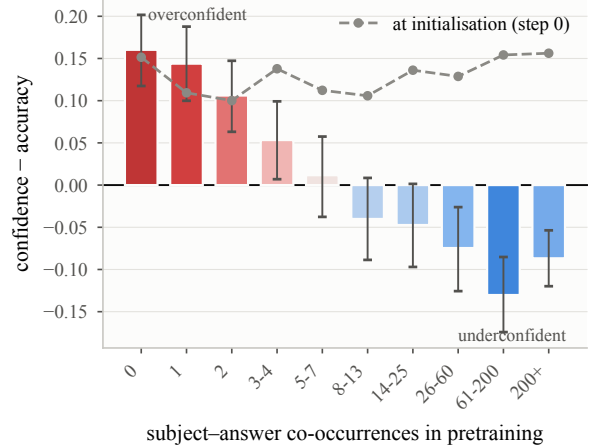


Figure 2: Confidence minus accuracy at the final checkpoint of the 1B six-epoch model. Error bars are bootstrap 95% intervals over facts. The dashed line is the same quantity at initialisation and is the reference the trained values should be read against, not zero.

Scoring	Gap by co-occurrence				ρ
	0	5–7	26–60	200+	
Length-norm.	0.160	0.012	-0.075	-0.087	-0.25
Summed	0.385	0.310	0.175	-0.009	-0.29
PMI	0.398	0.355	0.282	0.074	-0.21

Table 2: Confidence minus accuracy at the final checkpoint, at four representative bins, under each scoring function. ρ is the rank correlation between log co-occurrence count and the per-fact gap over all 2,993 facts. The level shifts and the sign is not preserved, but the decline across frequency is.

something, so we report the bins separately in Table 3: confidence falls slightly for rare facts and rises steeply for frequent ones, and the two movements cancel. Accuracy rises almost everywhere.

The dissociation is clearest in the middle bins. For facts with 8–13 co-occurrences, accuracy improves by 0.207 after step 1,000 while confidence moves by -0.020 . For the most frequent facts, confidence and accuracy rise together and stay close. Trajectories are in Figure 3.

4.3 Controls

Figure 4 repeats the final-checkpoint gap under three changes. Increasing model size from 170M to 1B leaves the rare-end gap essentially where it was, at 0.197, 0.158 and 0.160 respectively. Training the 1B model for one epoch instead of six gives a curve close to the six-epoch one, so repeated passes over the same corpus do not improve the match between confidence and accuracy. Two rephrasings of every template, one passive and one possessive,

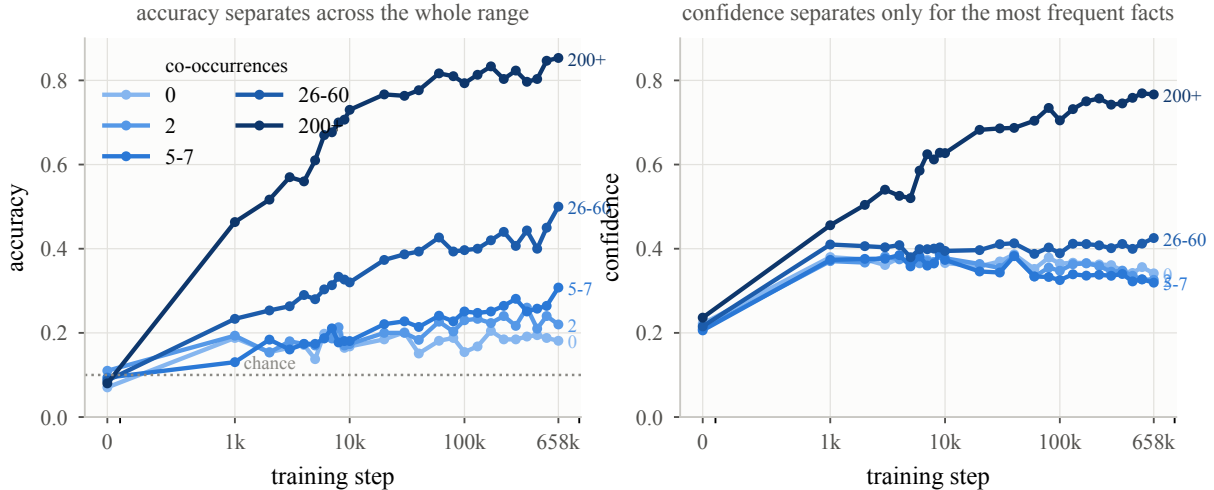


Figure 3: Accuracy and confidence over training for five representative bins. Accuracy separates across the whole frequency range; confidence separates only at the frequent end.

Co-occ.	Confidence		Accuracy	
	1K	final	1K	final
0	0.380	0.341	0.188	0.181
1	0.387	0.356	0.158	0.212
2	0.371	0.326	0.193	0.220
3-4	0.378	0.327	0.153	0.273
5-7	0.373	0.319	0.130	0.308
8-13	0.370	0.350	0.183	0.390
14-25	0.378	0.377	0.184	0.425
26-60	0.410	0.425	0.233	0.500
61-200	0.425	0.546	0.280	0.677
200+	0.456	0.767	0.463	0.853

Table 3: Confidence changes little after step 1,000 except at the frequent end, where it tracks a large accuracy gain. Accuracy improves across almost all bins.

reproduce the same shape, so the effect is not an artefact of our wording.

We compare the one-epoch and six-epoch models at their final checkpoints rather than at equal step counts, because the two use different learning-rate schedules and a matched-step comparison would confound the two factors.

4.4 Where the wrong answers come from

If confidence arrives before knowledge, something must be supplying the answers. We test whether wrong choices fall on the option the model would favour with no subject at all, scoring each candidate under the neutral prompt *The answer is* and taking the highest-scoring distractor as the model’s prior for that relation.

Among wrong answers, 52.0% land on that option, against 11.1% expected by chance ($p < 10^{-300}$, binomial). The rate peaks at 0.566 in the

Predictor	AUC	95% CI
Majority class	0.500	
Co-occurrence	0.784	[0.721, 0.842]
Probe confidence	0.868	[0.829, 0.906]
Both	0.852	[0.799, 0.899]

Table 4: Predicting free-generation errors on held-out entities, final checkpoint. “Both” is a logistic regression on the two features. The positive rate is 86.4%, so accuracy and F1 are uninformative here and are omitted. Intervals are bootstrapped over the 757 test facts.

middle of the range and falls to 0.357 in the most frequent bin (Figure 5). Using corpus frequency to define the prior instead of the model’s own preference gives the same trend at a lower rate. Errors on rare entities are therefore closer to a default guess for the relation, while errors on well-attested entities look more like specific confusions.

4.5 Predicting errors

We ask whether the quantities above predict failures on held-out entities. The label comes from free generation, namely whether the model produces a wrong answer, while the features come from the multiple-choice probe and from corpus counts, so no predictor has access to its own label.

Results are in Table 4. Co-occurrence count alone reaches 0.784 ROC-AUC and probe confidence alone reaches 0.868, both well above the majority-class baseline. Combining them changes little, and the difference of 0.016 has a bootstrap interval spanning zero. Because 86% of test facts are errors, accuracy and F1 are close to uninformative here.

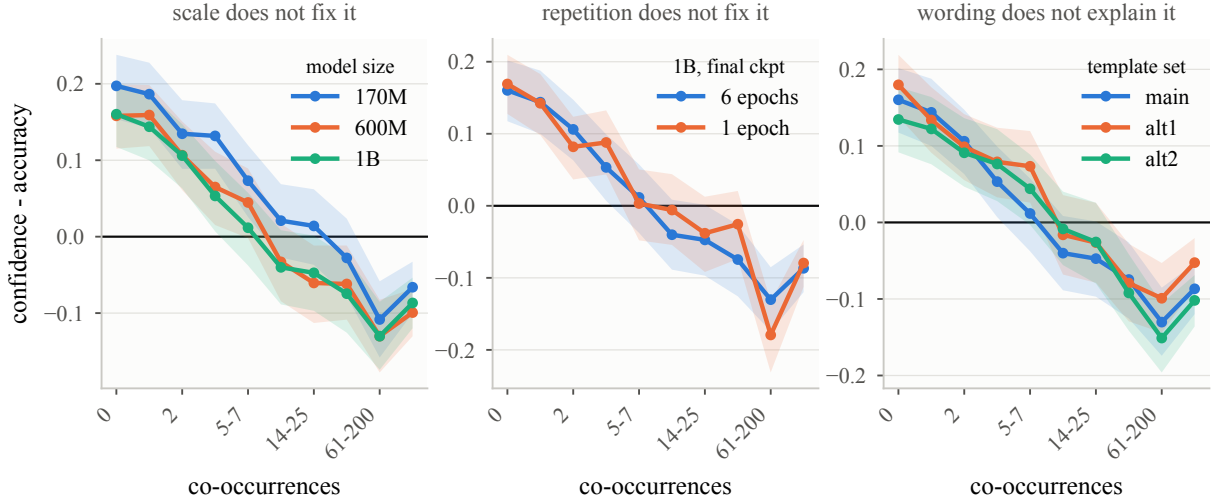


Figure 4: The gap under three controls. Model size, number of epochs and prompt wording all leave the pattern intact.

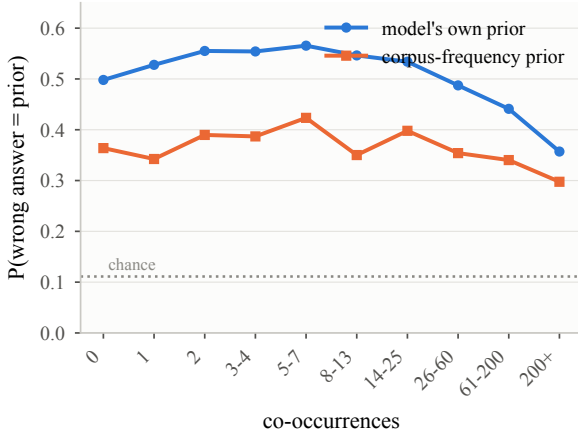


Figure 5: Share of wrong answers that match the model's preferred answer for the relation, measured with no subject in the prompt. Chance is 0.111.

The picture changes if we ask about confident errors rather than errors. Restricting to generations the model is sure about, the rate does not fall with frequency but peaks in the middle, at 0.360 for facts with 61–200 co-occurrences against 0.238 for facts with none, measured over all 2,993 facts rather than the held-out split. Bootstrap intervals over facts are 0.307 to 0.413 and 0.191 to 0.289, and the difference of 0.122 has an interval of 0.048 to 0.195. The ordering holds at the 50th, 75th and 90th percentiles of generation confidence. Producing a confidently wrong answer seems to need enough familiarity to construct a specific one. For the rarest entities the model tends to generate vague text instead.

5 Discussion

An untrained model is already overconfident, and uniformly so. What training supplies, and only where the corpus supports it, is the accuracy that would justify that confidence. Where it arrives the two align. Where it does not, the initial overconfidence simply stays.

The gap on rare entities is therefore a property of the training data rather than of a particular model. Nothing we varied removed it: a six-fold increase in parameters moved the rare-end gap by 0.037, and six passes over the corpus moved it less. [Kandpal et al. \(2023\)](#) estimate that reaching high accuracy on long-tail questions would need models many orders of magnitude larger than current ones. Over the range we tested, scale does little for long-tail calibration.

One result runs against the pattern and we do not have a confident explanation. The gap in the most frequent bin (-0.087) is smaller in magnitude than in the 61–200 bin (-0.130), so underconfidence peaks just below the top of the range rather than at it. This reversal appears in all four models we tested, so it is unlikely to be noise. A plausible reading is that at the highest frequencies the model is accurate enough that confidence starts catching up again, but we did not test this.

The two probing methods disagree. In multiple choice, where the model must distribute mass over a fixed set, overconfidence is largest for the rarest facts. In free generation, confident errors are most common in the middle of the frequency range. Both can be true: the multiple-choice set-

ting forces a commitment that free generation does not, and a model with no information about an entity produces hedged text rather than a confident falsehood. Which setting is the better model of deployed behaviour depends on the application.

6 Conclusion

Confidence and accuracy come apart during pre-training, and how far they come apart is predictable from a count that can be computed before training finishes. The separation is established within the first 1,000 of 658,032 steps, and survives changes in model size, epoch count and prompt wording. When the model does err on a rare entity, it falls back on its default answer for the relation more often than on anything specific to the entity asked about.

Limitations

Our models are small by current standards, at most 1B parameters, and are trained on Wikipedia alone. Whether the same pattern holds for models trained on mixed web corpora is untested. [Gottesman et al. \(2026\)](#) do report the same frequency-performance trend for models trained on broader data, but they score those models against their own Wikipedia-derived counts rather than counting the other corpora. What happens when the corpus itself changes is still open.

We can locate the onset of confidence no earlier than step 1,000, because that is the finest public checkpoint spacing, and it is available for only one of the four models. Most of the rise happens before that point, so we can say the separation is established early but not how early.

The ten-way multiple-choice probe is a convenient measurement, not a realistic task. It supplies a fixed candidate set that a deployed model would not have, and our free-generation results show the two settings do not always agree.

The lowest frequency bin is not a zero-knowledge control. Its accuracy is 0.181 against a chance level of 0.100, since the subject entity usually appears in the corpus even when it never co-occurs with the answer. That is not the whole story: on the 112 facts whose subject is absent entirely, accuracy is 0.205 [0.134, 0.286], which overlaps the bin rather than falling below it. The gap there is +0.124, so the pattern holds, but we cannot attribute the above-chance accuracy to subject exposure.

Co-occurrence count is correlated with how often the answer string appears on its own, so part of the trend we report may reflect familiarity with the answer rather than knowledge of the fact. Section 4.4 is consistent with that concern but does not settle it, and we did not run a probe that separates the two.

Finally, we cover 16 relations from one English dataset, and the probe set was generated from a single random seed. We report results for one seed rather than an average over several.

AI Disclosure and Reflection

We used Claude Code (Opus 5) as a helper throughout the project: for writing and debugging the notebook code, including the parts using HuggingFace and PyTorch libraries we had not worked with before, for producing the figures, for drafting and tidying the prose, and for discussing how to read the results.

The main lesson was that its output needs checking rather than trusting. It was reliable on code and library details, and least reliable with numbers. Several values were attributed to the wrong frequency bin, and quantities quoted in the text did not always match the tables they came from. We caught these by recomputing everything from the saved per-fact results and reading the tables against the text.

Acknowledgments

We thank the authors of LMEnt for releasing the models, corpus annotations and checkpoint metadata that this project depends on.

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